

Abdelrahman Sayed Sayed Emam Ibrahim

Current position: Marie Skłodowska-Curie PhD Fellow @ ESTAS lab Université Gustave Eiffel

Contact information

Laboratoire ESTAS

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[Homepage](#)

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Research interests

- Safety verification of neural networks and AI controlled systems
- Hybrid control, formal methods, reachability analysis, metaheuristic optimization
- Safe autonomous subsea & railway operations
- Applications: autonomous underwater vehicles, multi-agent systems, robotic manipulators

Education

11/2023-Present	Université Gustave Eiffel , Lille, France Marie Skłodowska-Curie Actions (MSCA) CO-FUNDED PhD thesis in Computer Science and Automatic Control (acceptance rate: 7.7%) <i>Supervisors:</i> Pierre-Jean Meyer and Mohamed Ghazel (UGE, France), Asgeir Johan Sørensen (NTNU, Norway) <i>Thesis:</i> Formal verification of neural ODE for safety evaluation in autonomous vehicles
10/2021-10/2023	Erasmus Mundus Joint Master's Degree in Marine and Maritime Intelligent Robotics (MIR) (acceptance rate: 6.1%) Master's in Engineering of Complex Systems, Université de Toulon , France Master's in Marine Robotics, Department of Marine Technology-NTNU , Norway <i>Supervisors:</i> Asgeir Johan Sørensen and Ingrid Bouwer Utne <i>Thesis:</i> Risk Aware Control of Underwater Snake Robots in Confined Environments
09/2015-02/2020	Nile University , Giza, Egypt Bachelor of Engineering and Applied Sciences in Mechanical Engineering <i>Supervisors:</i> Hossam Hassan Ammar and Raafat Shalaby <i>Thesis:</i> Design and Control of an 18 DOF Hexapod Multi-agent Swarm for Search and Rescue Missions

Professional experience

01/2026-Present	Visiting Researcher , AUR & MC Labs, NTNU, Trondheim, Norway <i>Supervisor:</i> Asgeir J.Sørensen
06/2023-09/2023	Research Intern , ESTAS, Université Gustave Eiffel, Lille, France <i>Supervisor:</i> Pierre-Jean Meyer <i>Topic :</i> Bridging discrete and continuous neural network models
01/2023–04/2023	Graduate Teaching Assistant , Department of Marine Technology - NTNU, Trondheim, Norway
06/2023-08/2022	Summer Research Intern , LIRMM - Explore team, Université de Montpellier, Montpellier, France <i>Supervisor:</i> Lionel Lapierre <i>Topic :</i> Conception of a poly-articulated underwater robot for karstic exploration
04/2021–08/2021	Robotics Engineer , Plumake srl and Verona FabLab, Verona, Italy
03/2020–02/2021	STEM Curriculum Developer , STEM HUB, Dubai, UAE
10/2019–02/2020	Undergraduate Research Assistant , Smart Engineering Systems Research Center (SESC) - Nile University, Giza, Egypt <i>Supervisors :</i> Ahmad Taher Azar, Hossam Hassan Ammar and Raafat Shalaby
07/2019–08/2019	Maker’s Trainer , Arab Digital Expression Foundation, Cairo, Egypt
02/2019–06/2019	Control Engineer , Erasmus+KA2 VET-ENG project - Nile University , Giza, Egypt
09/2018–02/2019	Junior Teaching Assistant , Nile University, Giza, Egypt

Awards

June 2025	Best Poster at Doctoral Day 2025 - MADIS & ENGSYS Graduate Schools Awarding institution : Université de Lille
September 2023	Second Place at NTNU Faculty of Engineering Innovation Competition for Master thesis Awarding institution : Faculty of Engineering, Norwegian University of Science and Technology (NTNU) Awarded : 20,000 NOK
June 2023	Best Student Presentation at Underwater Technology Conference “UTC 2023” Awarding institution : Underwater Technology Foundation Awarded : 10,000 NOK
November 2020	Best Undergraduate Research Thesis for the academic year 2019 - 2020 Awarding institution : School of Engineering and Applied Sciences, Nile University
2018- 2019	Dean’s Honor list for the following semesters (SP’18, FA’18, SP’19) Awarding institution : Nile University
October 2016	4th Place in Minesweepers International competition Awarding institution : Zewail City of Science and Technology
October 2016	No Pain No Gain special prize in Minesweepers International competition Awarding institution : Zewail City of Science and Technology

Grants & Project calls

- December 2024 **Mobility grant** from MADIS Doctoral School for 2025-2026
Awarding institution : Mathematics and Digital Sciences (MADIS) Graduate school
Awarded : **€1000**
- October 2019 **Bachelor Thesis grant** for the academic year 2019 - 2020
Awarding institutions : Graduation Projects Funding Program by Egyptian Academy of Scientific Research and Technology (ASRT) and Nile University (NU)
Awarded : **€3380: €2340** from ASRT and **€1040** from NU

Travel awards & grants

- November 2025 **Travel award** from AAAI for AAAI 2026
Awarding institution : Association for the Advancement of Artificial Intelligence (AAAI)
Awarded : **\$2000**
- July 2024 **Travel grant** for IGSC M18: Control and Machine Learning
Awarding institution : EECI-European Embedded Control Institute
Awarded : **€400**
- August 2023 **Travel grant** for SCIoI & ISAB Summer School
Awarding institution : Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy – EXC-2002/1 – Project number 390523135
Awarded : **€400**
- June 2023 **Travel grant** for OCEANS 2023 conference
Awarding institution : Erasmus+ Programme of the European Union
Awarded : **€500**

Teaching

Graduate Teaching assistant

NTNU, Trondheim, Norway

- *Modelling, Simulation and Analysis of Mechatronic Systems* (TMR4275), Department of Marine Technology (IMT), 2023

Junior Teaching assistant

Nile University, Giza, Egypt

- *Numerical Methods for Engineers* (MATH303), School of Engineering and Applied Sciences (EAS), 2018-2019

Mentoring

11/2025-Present	Yahia Salaheldin Shaaban , Master student at Mohamed bin Zayed University of Artificial Intelligence <i>Project:</i> Extending KKL observers to non-autonomous nonlinear systems using hypernetworks <i>Role:</i> Mentor
06/2023-06/2024	Ramy Antar Ahmed Alham , Erasmus Mundus MIR Master's student at Université de Toulon and NTNU. <i>Project:</i> Enhancing the Navigation and Fault Detection in Small Remotely Operated Vehicles (ROVs) <i>Role:</i> Mentor

Service

Administration

2025	Elected representative of PhD students in the council of the Mathematics and Digital Sciences (MADIS) Graduate school, Université de Lille, France
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Professional Memberships

2025-Present	Association for the Advancement of Artificial Intelligence (AAAI) Student Silver Technical Member
2023-Present	Marine Technology Society (MTS) Student Member
2023-Present	Marie Curie Alumni Association (MCAA)
2021-Present	Erasmus Mundus Alumni Association (EMA)

Reviewing

Conferences. Epistemic Intelligence in Machine Learning workshop ([EIML@ICML 2026](#)), Weight-Space Symmetries: from Foundations to Practical Applications workshop ([WSS@ICML 2026](#)), Trustworthy AI for Good workshop ([AI4GOOD@ICML 2026](#)), International Conference on Computer Aided Verification Artifact Evaluation Committee ([CAV 2026](#)), Post-AGI Science and Society Workshop ([P-AGI@ICLR 2026](#)), Workshop on Mathematical Reasoning and AI ([MATH-AI@NeurIPS 2025](#)), Symmetry and Geometry in Neural Representations workshop ([NeurReps@NeurIPS 2025](#)), Unifying Representations in Neural Models workshop ([UniReps@NeurIPS 2025](#)), Dynamics at the Frontiers of Optimization, Sampling, and Games workshop ([DynaFront@NeurIPS 2025](#)), International Workshop on Artificial Intelligence for RAILwayS ([AI4RAILS](#)), IEEE Novel Intelligent and Leading Emerging Sciences conference (NILES)

Invited seminar and workshop talks

March 2026	<i>Safety Verification of Neural ODE based perception modules on AUVs</i> NTNU-Department of Marine Technology PhD Seminar Series, Trondheim, Norway
December 2025	<i>Safety Verification of Neural ODE models</i> Université Gustave Eiffel Séminaire ESTAS, Villeneuve d'Ascq, France

October 2025	<i>Bridging Neural ODE and ResNet</i> Workshop on cautiousness and imprecision in machine learning (CIML 2025) at Sorbonne Université, Paris, France
October 2025	<i>Bridging Neural ODE and ResNet</i> Université Gustave Eiffel AI seminar, Marne-la-Vallée, France

Summer & Winter Schools

February 2025	Winter School on Numerical Methods for Optimal Control of Nonsmooth Systems Mines Paris-PSL, Paris, France
July 2024	IGSC: International Graduate School on Control - M18: Control and Machine Learning University of Dubrovnik, Dubrovnik, Croatia
February 2024	Set4Most: Modelling, Estimation, and Control using Set-Based Methods Centrale Lille, Villeneuve d'Ascq, France
October 2023	SAUTOS: Safe AUTOnomous Systems Université Paris-Saclay, Orsay, France
August 2023	SCIoI & ISAB Summer School Science of Intelligence (SCIoI), Berlin, Germany

Languages

Arabic	Native
English	Fluent (IELTS score 7)
French	Professional working proficiency (B2)
Italian	Limited working proficiency (A2)

Publications

[Google Scholar page](#) — [ResearchGate](#) — [ORCiD](#) — [OpenReview](#)

References

PhD Thesis Supervisors

1. [Pierre-Jean Meyer](#)
 - Relation: PhD Thesis main supervisor
 - Email: pierre-jean.meyer-AT-univ-eiffel.fr
2. [Mohamed Ghazel](#)
 - Relation: PhD Thesis director
 - Email: mohamed.ghazel-AT-univ-eiffel.fr
3. [Asgeir Johan Sørensen](#)
 - Relation: PhD Thesis co-supervisor & Master Thesis main supervisor
 - Email: asgeir.sorensen-AT-ntnu.no

Master Thesis Supervisors

1. **Ingrid Bouwer Utne**

- Relation: Master Thesis co-supervisor
- Email: ingrid.b.utne-AT-ntnu.no

2. **Lionel Lapierre**

- Relation: Master Thesis Internship supervisor
- Email: lionel.lapierre-AT-ensta-bretagne.fr

3. **Vincent Hugel**

- Relation: MIR Master's Head of Studies & Main supervisor for master's 1st year
- Email: vincent.hugel-AT-univ-tln.fr

Bachelor Thesis Supervisors

1. **Hossam Hassan Ammar**

- Relation: Bachelor Thesis main supervisor
- Email: h.ammar-AT-gaf.edu.eg

2. **Raafat Shalaby**

- Relation: Bachelor Thesis director
- Email: rshalaby-AT-nu.edu.eg

3. **Ahmad Taher Azar**

- Relation: Bachelor Research supervisor
- Email: ahmad.t.azar-AT-ieee.org